ESMA 6661: Assignment 2

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Problem 1: Definitions

(a) **Sufficiency principle:** The sufficient principle tell that a parameter is enough to explain the population with the data.

(b) **Factorization theorem:** It is a tetanic where you separate the pdf in a way such that you have the product of 2 function for all sample points x and all parameter points θ

$$f(x|\theta) = g(T(x)|\theta)h(x).$$

(c) **Likelihood Function:** It is the probability function that tells how well a parameter preforms given a dataset

(d) **Minimal sufficient:** it is the minimum data required to explain the parameter but still retaining all of the information

(e) **Complete statistics:** It is a complete information of the estimator without depending in outside factors.

Problem 2: Sufficient Statistic

In each of these exercises, assume that the random variables $X_1, \ldots, X_n$ form a random sample of size n from the distribution specified in that exercise, and show that the statistic T specified in the exercise is a sufficient statistic for the parameter.

(a) The Bernoulli distribution with parameter p, which is unknown $(0 < p < 1); T = \sum_n^{i=1} X_i$.

$$P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}, \quad x_i \in \{0, 1\}.$$

$$f(x_1, \ldots, x_n; p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}.$$

$$f(x_1, \ldots, x_n; p) = p^{\sum_{i=1}^{n} x_i}(1-p)^{\sum_{i=1}^{n}(1-x_i)}.$$

$$\sum_{i=1}^{n}(1-x_i) = n - \sum_{i=1}^{n} x_i.$$

$$f(x_1, \ldots, x_n; p) = p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i}.$$

Let

$$T = \sum_{i=1}^{n} x_i.$$

Then

$$f(x_1, \ldots, x_n; p) = p^T(1-p)^{n-T}.$$

We can write this as

$$f(x_1, \ldots, x_n; p) = \underbrace{p^T(1-p)^{n-T}}_{g(T,p)} \cdot \underbrace{1}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T = \sum_{i=1}^{n} X_i \text{ is a sufficient statistic for } p.}$$

(b) The geometric distribution with parameter p, which is unknown $(0 < p < 1); T = \sum_n^{i=1} X_i$.

$$P(X_i = x_i) = p(1-p)^{x_i - 1}, \quad x_i = 1, 2, 3, \ldots$$

$$f(x_1, \ldots, x_n; p) = \prod_{i=1}^{n} p(1-p)^{x_i - 1}.$$

$$f(x_1, \ldots, x_n; p) = p^n (1-p)^{\sum_{i=1}^{n} (x_i - 1)}.$$

$$\sum_{i=1}^{n} (x_i - 1) = \sum_{i=1}^{n} x_i - n.$$

$$f(x_1, \ldots, x_n; p) = p^n (1-p)^{\sum_{i=1}^{n} x_i - n}.$$

$$T = \sum_{i=1}^{n} x_i.$$

$$f(x_1, \ldots, x_n; p) = p^n (1-p)^{T-n}.$$

$$f(x_1, \ldots, x_n; p) = \underbrace{p^n (1-p)^{T-n}}_{g(T,p)} \cdot \underbrace{1}_{h(x_1, \ldots, x_n)}.$$

$$\boxed{T = \sum_{i=1}^{n} X_i \text{ is a sufficient statistic for } p.}$$

(c) The negative binomial distribution with parameters r and p, where r is known and p is unknown $(0 < p < 1); T = \sum_n^{i=1} X_i$.

$$P(X_i = x_i) = \binom{x_i - 1}{r - 1} p^r (1-p)^{x_i - r}, \quad x_i = r, r+1, r+2, \ldots$$

$$f(x_1, \ldots, x_n; p) = \prod_{i=1}^{n} \binom{x_i - 1}{r - 1} p^r (1-p)^{x_i - r}.$$

$$f(x_1, \ldots, x_n; p) = \left(\prod_{i=1}^{n} \binom{x_i - 1}{r - 1} \right) p^{nr} (1-p)^{\sum_{i=1}^{n} (x_i - r)}.$$

$$\sum_{i=1}^{n} (x_i - r) = \sum_{i=1}^{n} x_i - nr.$$

$$f(x_1, \ldots, x_n; p) = \left(\prod_{i=1}^{n} \binom{x_i - 1}{r - 1} \right) p^{nr} (1-p)^{\sum_{i=1}^{n} x_i - nr}.$$

$$T = \sum_{i=1}^{n} x_i.$$

$$f(x_1, \ldots, x_n; p) = \underbrace{p^{nr} (1-p)^{T-nr}}_{g(T,p)} \underbrace{\left(\prod_{i=1}^{n} \binom{x_i - 1}{r - 1} \right)}_{h(x_1, \ldots, x_n)}.$$

$$\boxed{T = \sum_{i=1}^{n} X_i \text{ is a sufficient statistic for } p.}$$

(d) The normal distribution for which the mean μ is known and the variance $\sigma^2 > 0$ is unknown; $T = \sum_n^{i=1}(X_i - \mu)^2$.

$$T = \sum_{i=1}^{n}(X_i - \mu)^2$$

$$f(x_i; \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right), \quad -\infty < x_i < \infty.$$

$$f(x_1, \ldots, x_n; \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right).$$

$$T = \sum_{i=1}^{n}(x_i - \mu)^2.$$

$$f(x_1, \ldots, x_n; \sigma^2) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{T}{2\sigma^2}\right)}_{g(T,\sigma^2)} \cdot \underbrace{1}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T = \sum_{i=1}^{n}(X_i - \mu)^2 \text{ is a sufficient statistic for } \sigma^2.}$$

(e) The gamma distribution with parameters α and β, where the value of α is known and the value of β is unknown $(\beta > 0); T = \bar{X}_n$. To show that

$$T = \bar{X}_n$$

$$f(x_i; \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} \exp\left(-\frac{x_i}{\beta}\right), \quad x_i > 0.$$

$$f(x_1, \ldots, x_n; \beta) = \prod_{i=1}^{n} \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} \exp\left(-\frac{x_i}{\beta}\right).$$

$$f(x_1, \ldots, x_n; \beta) = \frac{1}{\Gamma(\alpha)^n \beta^{n\alpha}} \left(\prod_{i=1}^{n} x_i^{\alpha-1}\right) \exp\left(-\frac{\sum_{i=1}^{n} x_i}{\beta}\right).$$

$$T = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

$$\sum_{i=1}^{n} x_i = n\bar{X}_n = nT,$$

$$f(x_1, \ldots, x_n; \beta) = \underbrace{\beta^{-n\alpha} \exp\left(-\frac{nT}{\beta}\right)}_{g(T,\beta)} \cdot \underbrace{\frac{1}{\Gamma(\alpha)^n} \prod_{i=1}^{n} x_i^{\alpha-1}}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T = \bar{X}_n \text{ is a sufficient statistic for } \beta.}$$

(f) The gamma distribution with parameters α and β, where the value of β is known and the value of β is unknown $(\beta > 0); T = \prod_{i=1}^{n} X_n$.

$$T = \prod_{i=1}^{n} X_i$$

$$f(x_i; \alpha) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} \exp\left(-\frac{x_i}{\beta}\right), \quad x_i > 0.$$

$$f(x_1,\ldots,x_n;\alpha)=\prod_{i=1}^{n}\frac{1}{\Gamma(\alpha)\beta^{\alpha}}x_i^{\alpha-1}\exp\left(-\frac{x_i}{\beta}\right).$$

$$f(x_1,\ldots,x_n;\alpha)=\frac{1}{\Gamma(\alpha)^n\beta^{n\alpha}}\left(\prod_{i=1}^{n}x_i^{\alpha-1}\right)\exp\left(-\frac{\sum_{i=1}^{n}x_i}{\beta}\right).$$

$$\prod_{i=1}^{n}x_i^{\alpha-1}=\left(\prod_{i=1}^{n}x_i\right)^{\alpha-1}.$$

$$T=\prod_{i=1}^{n}X_i.$$

$$f(x_1,\ldots,x_n;\alpha)=\underbrace{\Gamma(\alpha)^{-n}\beta^{-n\alpha}T^{\alpha-1}}_{g(T,\alpha)}\cdot\underbrace{\exp\left(-\frac{\sum_{i=1}^{n}x_i}{\beta}\right)}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T=\prod_{i=1}^{n}X_i \text{ is a sufficient statistic for } \alpha.}$$

(g) The beta distribution with parameters α and β, where the value of β is known and the value of β is unknown $(\beta>0); T=\prod_{i=1}^{n}X_n$.

$$T=\prod_{i=1}^{n}X_i$$

$$f(x_i;\alpha)=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x_i^{\alpha-1}(1-x_i)^{\beta-1},\quad 0<x_i<1.$$

$$f(x_1,\ldots,x_n;\alpha)=\prod_{i=1}^{n}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x_i^{\alpha-1}(1-x_i)^{\beta-1}.$$

$$f(x_1,\ldots,x_n;\alpha)=\left[\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\right]^n\left(\prod_{i=1}^{n}x_i^{\alpha-1}\right)\left(\prod_{i=1}^{n}(1-x_i)^{\beta-1}\right).$$

$$\prod_{i=1}^{n}x_i^{\alpha-1}=\left(\prod_{i=1}^{n}x_i\right)^{\alpha-1}.$$

$$T=\prod_{i=1}^{n}X_i.$$

$$f(x_1,\ldots,x_n;\alpha)=\underbrace{\left[\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\right]^n T^{\alpha-1}}_{g(T,\alpha)}\cdot\underbrace{\prod_{i=1}^{n}(1-x_i)^{\beta-1}}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T=\prod_{i=1}^{n}X_i \text{ is a sufficient statistic for } \alpha.}$$

(h) The discrete uniform distribution take values of $1,2,\ldots,\theta$,

$$f(x|\theta)=\frac{1}{\theta}; x\in\{1,2,\ldots,\theta\}$$

$$f(x_1,\ldots,x_n\mid\theta)=\prod_{i=1}^{n}\frac{1}{\theta}=\theta^{-n},\quad \text{if } 1\le x_i\le\theta \text{ for all } i.$$

$$f(x_1, \ldots, x_n \mid \theta) = \theta^{-n} \mathbf{1}_{\{\max(x_1,\ldots,x_n) \leq \theta\}},$$
$$T = \max(X_1, \ldots, X_n).$$
$$f(x_1, \ldots, x_n \mid \theta) = \underbrace{\theta^{-n} \mathbf{1}_{\{T \leq \theta\}}}_{g(T,\theta)} \cdot \underbrace{1}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T = \max(X_1, \ldots, X_n) \text{ is a sufficient statistic for } \theta.}$$

(i) The uniform distribution on the interval $[a, b]$, where the value of b is known and the value of b is unknown $(a > b); T = max\{X_1, \ldots, X_n\}$.

$$f(x \mid a) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

$$f(x_1, \ldots, x_n \mid a) = \prod_{i=1}^{n} \frac{1}{b-a} = (b-a)^{-n}, \quad \text{if } a \leq x_i \leq b \text{ for all } i.$$
$$f(x_1, \ldots, x_n \mid a) = (b-a)^{-n} \mathbf{1}_{\{a \leq \min(x_1,\ldots,x_n)\}}.$$
$$T = \min(X_1, \ldots, X_n).$$
$$f(x_1, \ldots, x_n \mid a) = \underbrace{(b-a)^{-n} \mathbf{1}_{\{a \leq T\}}}_{g(T,a)} \cdot \underbrace{1}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T = \min(X_1, \ldots, X_n) \text{ is a sufficient statistic for } a.}$$

(j) The uniform distribution on the interval $[a, b]$, where the value of b is known and the value of b is unknown $(a < b); T = min\{X_1, \ldots, X_n\}$. Let $X_1, \ldots, X_n$ be a random sample from $U[a, b]$, with pdf

$$f(x \mid b) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

$$f(x_1, \ldots, x_n \mid b) = \prod_{i=1}^{n} \frac{1}{b-a} = (b-a)^{-n}, \quad \text{if } a \leq x_i \leq b \text{ for all } i.$$
$$f(x_1, \ldots, x_n \mid b) = (b-a)^{-n} \mathbf{1}_{\{\max(x_1,\ldots,x_n) \leq b\}}.$$
$$T = \max(X_1, \ldots, X_n).$$
$$f(x_1, \ldots, x_n \mid b) = \underbrace{(b-a)^{-n} \mathbf{1}_{\{T \leq b\}}}_{g(T,b)} \cdot \underbrace{1}_{h(x_1,\ldots,x_n)}.$$

$$\boxed{T = \max(X_1, \ldots, X_n) \text{ is a sufficient statistic for } b.}$$

Problem 3: Sufficient problem

Let $X \sim N(0, \sigma^2)$ with $\sigma^2 > 0$ unknown. Determine if $|X|$ is a sufficient.

$$f(x; \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

Since $x^2 = |x|^2$, we can write:

$$f(x; \sigma^2) = \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{|x|^2}{2\sigma^2}\right)}_{g(|x|,\sigma^2)} \cdot \underbrace{1}_{h(x)}$$

Problem 4: Random Variables

Let $X_1, \ldots, X_n$ be independent random variables with densities.

$$f_{x_i}(x|\theta) = exp\{i\theta - x\}I(x \geq i\theta);$$

where $I(x \geq i\theta)$ is the indicator function,

$$I(x \geq i\theta) = \begin{cases} 1, & x \geq i\theta \\ 0, & x < i\theta \end{cases}$$

Prove that $T = min(X_i/i)$ is a sufficient statistic for θ.

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i \mid \theta) = \prod_{i=1}^{n} e^{i\theta - x_i}\, I(x_i \geq i\theta).$$

$$L(\theta; x_1, \ldots, x_n) = \exp\left(\theta \sum_{i=1}^{n} i - \sum_{i=1}^{n} x_i\right) \prod_{i=1}^{n} I(x_i \geq i\theta).$$

$$\prod_{i=1}^{n} I(x_i \geq i\theta) = I\left(\theta \leq \min_{i=1,\ldots,n} \frac{x_i}{i}\right).$$

$$L(\theta; x_1, \ldots, x_n) = \exp\left(\theta \sum_{i=1}^{n} i - \sum_{i=1}^{n} x_i\right) I\left(\theta \leq \min_{i=1,\ldots,n} \frac{x_i}{i}\right).$$

$$\hat{\theta}_{\text{MLE}} = \min_{i=1,\ldots,n} \frac{X_i}{i}.$$

Problem 5: TBD

For each of the following distribution let $X_1, \ldots, X_n$, be a random sample find a minimal sufficient statistic for θ

1. **Normal:**

$$f(x|\theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}, \quad -\infty < x < \infty; -\infty < \theta < \infty$$

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i - \theta)^2\right).$$

$$\sum_{i=1}^{n}(x_i - \theta)^2 = \sum_{i=1}^{n} x_i^2 - 2\theta \sum_{i=1}^{n} x_i + n\theta^2$$

$$L(\theta; x_1, \ldots, x_n) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{n} x_i^2\right) \cdot \exp\left(\theta \sum_{i=1}^{n} x_i - \frac{n\theta^2}{2}\right).$$

$$L(\theta; x_1, \ldots, x_n) = h(x_1, \ldots, x_n)\, g\Big(T(x_1, \ldots, x_n), \theta\Big)$$

$$\boxed{T(X_1, \ldots, X_n) = \sum_{i=1}^{n} X_i}$$

2. **Location exponential:**

$$f(x|\theta) = e^{-(x-\theta)}, \quad \theta < x < \infty, \quad -\infty < \theta < \infty$$

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = \prod_{i=1}^{n} e^{-(x_i-\theta)} \mathbf{1}_{\{x_i > \theta\}} = e^{-\sum_{i=1}^{n}(x_i-\theta)} \mathbf{1}_{\{\theta < X_{(1)}\}},$$

$$L(\theta; x_1, \ldots, x_n) = e^{n\theta - \sum_{i=1}^{n} x_i} \mathbf{1}_{\{\theta < X_{(1)}\}}.$$

$$L(\theta; x_1, \ldots, x_n) = \underbrace{e^{-\sum x_i}}_{h(x_1,\ldots,x_n)} \cdot \underbrace{e^{n\theta} \mathbf{1}_{\{\theta < X_{(1)}\}}}_{g(X_{(1)},\theta)}.$$

$$\boxed{T(X_1, \ldots, X_n) = X_{(1)} = \min(X_1, \ldots, X_n)}$$

3. **Logistic:**

$$f(x|\theta) = \frac{e^{-(x-\theta)}}{(1 + e^{-(x-\theta)})^2}, \quad -\infty < x < \infty; -\infty < \theta < \infty$$

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = \prod_{i=1}^{n} \frac{e^{-(x_i-\theta)}}{\left(1 + e^{-(x_i-\theta)}\right)^2} = e^{n\theta - \sum_{i=1}^{n} x_i} \prod_{i=1}^{n} \frac{1}{\left(1 + e^{-(x_i-\theta)}\right)^2}.$$

$$\boxed{T(X_1, \ldots, X_n) = (X_{(1)}, X_{(2)}, \ldots, X_{(n)})}.$$

4. **Cauchy:**

$$f(x|\theta) = \frac{1}{\pi} \frac{1}{1 + (x-\theta)^2}; \quad -\infty < x < \infty; \infty < \theta < \infty$$

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = \prod_{i=1}^{n} \frac{1}{\pi} \frac{1}{1 + (x_i-\theta)^2} = \frac{1}{\pi^n} \prod_{i=1}^{n} \frac{1}{1 + (x_i-\theta)^2}.$$

$$\boxed{T(X_1, \ldots, X_n) = (X_{(1)}, X_{(2)}, \ldots, X_{(n)})}.$$

5. **Laplace (double exponential):**

$$f(x|\theta) = \frac{1}{2} e^{-|x-\theta|}, \quad -\infty < x < \infty; -\infty < \theta < \infty$$

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = \prod_{i=1}^{n} \frac{1}{2} e^{-|x_i-\theta|} = \frac{1}{2^n} e^{-\sum_{i=1}^{n} |x_i-\theta|}.$$

$$T(X_1, \ldots, X_n) = (X_{(1)}, X_{(2)}, \ldots, X_{(n)}),$$

$$\boxed{T(X_1, \ldots, X_n) = (X_{(1)}, X_{(2)}, \ldots, X_{(n)}) \text{ (order statistics)}}.$$

Problem 6: TBD

Let $X_1, \ldots, X_n$ be i.i.d with geometric distribution

$$P_\theta(X = x) = \theta(1-\theta)^{x-1}, \quad x = 1, 2, \ldots, \quad 0 < \theta < 1$$

Show that $\sum X_i$ is sufficient for θ, and find the family of distributions of $\sum X_i$. Is the family complete?

Problem 7: TBD

Let X be one observation from the pdf

$$f(x|\theta) = \left(\frac{\theta}{2}\right)^{|x|}(1-\theta), \quad x = -1, 0, 1, \quad 0 < \theta < 1$$

1. Is X a complete sufficient statistic?
2. Is $|X|$ a complete sufficient statistic?
3. Does $f(\dot{|}\theta)$ belong to the exponential class?

Problem 8: TBD

Another popular distribution is the inverse Gaussian distribution. This distribution has many applications, particularly in modeling of lifetimes. See the books by Chikhris and Folks (1989) and Seshadri (1993). Let $X_1, \ldots, X_n$ be a random sample from the inverse Gaussian distribution with pdf

$$f(x|\mu, \lambda) = \frac{\lambda}{\sqrt{2\pi X^3}} \exp\left(-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right), 0 < x < \infty$$

(a) Show that the statistics

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and } T = \frac{n}{\sum_{i=n}^{n} \frac{1}{X_i} - \frac{1}{X}}$$

are sufficient and complete

(b) For part 2, show that $\bar{X}$ has an inverse Gaussian distribution, and $n\lambda/T$ has a χ_1^2 distribution, and they are independent. (Schwarz and Samanta 1991 do the general case.)